GENESIS OS / RELATIONSHIP SCORES

UNIFIED MASTER ENGINEERING BLUEPRINT

Developer Build Contract — Clean, Corrected, Workbook-Aligned Revision

Status: build-governing revision
Revision date: March 28, 2026
Primary stance: deterministic workbook law first; governed runtime layers second; AI explanation and extraction third

This revision locks	What changed from the merged source file
One canonical compatibility formula	The workbook-aligned 52 / 23 / 13 / 12 pair law is the only official production formula in this document.
One clear product doctrine	Compatibility remains the match-ranking score. Readiness and Relationship Forecast are separate versioned products and do not replace compatibility.
One counting convention	Sheet dimensions count the header row. Data-row counts exclude the header row. This resolves the 48,205 vs 48,206 style conflicts.
One source-of-truth order	Workbook wins on score law. This blueprint wins on implementation narrative, runtime boundaries, storage, and build order.
One governance stance	Companion modules, clusters, drift, Sanctuary, Quantum Mirror, and AI can enrich evidence and explanation; they do not rewrite core score law.

Document control

Field	Value
Purpose	Give engineering, product, data, AI, QA, and future technical partners one clean implementation document with the contradictions removed and the source-of-truth order locked.
Authoritative upstream source	genesis_os_final_quantum_scoring_workbook_v10_polarity.xlsx (workbook law assumed at build time)
This document supersedes	Legacy score formulas, duplicated blueprint fragments, and any merged section that conflicts with the workbook-aligned pair law or the doctrine locked here.
Primary audience	Founder, product, backend, Bubble/full-stack, AI, ML, analytics, QA, and implementation partners.
Recommended handoff bundle	This blueprint + workbook + import scripts + MatchHarness fixtures + schema registry + cluster rule packs.
Non-goal	This document is not a marketing deck and not a vague product manifesto. It is build law.

Contents

- Part I — Governance, corrections, and source-of-truth rules
- Part II — Canonical scoring engine and workbook import law
- Part III — Readiness, forecast, clusters, drift, and shadow activation
- Part IV — Runtime architecture, storage, APIs, and eventing
- Part V — Sanctuary, Quantum Mirror, relationship mode, and governed enrichment
- Part VI — QA, rollout, schema inventory, and implementation warnings
- Appendices — controlled labels, cluster catalog, module registry, deprecated items, release checklist

Part I — Governance, corrections, and source-of-truth rules

1. Executive lock summary

This blueprint is the clean developer version of the Genesis OS package. It keeps the right parts of the uploaded master file, removes duplicated doctrine, resolves score-law conflicts, and turns the merged package into one coherent build contract.

The uploaded file carried multiple document generations inside one shell: a workbook-aligned master blueprint, non-scoring extracts from the older master blueprint, companion-system contracts, missing-data layers, readiness and forecast contracts, cluster packs, Sanctuary contracts, Quantum Mirror contracts, schema dictionaries, and later addenda. That material is valuable, but left ungoverned it causes spec drift. This document fixes that.

- CompatibilityScore is locked to the workbook-aligned pair-quantum law and only that law.
- Compatibility is the primary ranking score for candidate matching and access gating.
- Readiness and Relationship Forecast are real products, but they are separate contracts, separate formulas, and separate exposure rules.
- Sheet-dimension counts and data-row counts are now explicitly distinguished so developers stop hard-coding disputed constants.
- Companion systems, clusters, drift, shadow activation, Sanctuary, Quantum Mirror, and ML are officially inside the package, but only as governed layers around the core engine.
- Premium, economy, badges, vanity identity, and social layers are hard-forbidden from changing rank, score, eligibility, or confidence.

2. Source-of-truth order

Priority	Artifact	What it governs	Rule
1	Workbook	Score law, matrices, pair quantum metrics, overlays, polarity axes, tracked scoring fields	If a formula or tracked scoring metric conflicts with anything else, the workbook wins.
2	This unified blueprint	Implementation narrative, service ownership, storage, runtime order, exposure rules, build order	If legacy Word text conflicts with this document, this document wins.
3	Specialized governed contracts	Drift, shadow activation, Sanctuary, Quantum Mirror, readiness/forecast, schema, cluster packs	These extend the system only where they do not override workbook pair law or this blueprint.
4	Legacy blueprint text	Non-scoring product ideas, historical context, older UX concepts	Use only when it still matches items 1–3. Legacy score math is deprecated.

3. Corrections this revision makes on purpose

Conflict in merged source	Decision in this revision	Developer instruction
Workbook-aligned pair formula vs alternate '0.30*B + 0.22*A + ...' production formula	The alternate production formula is deprecated.	Build only the workbook-aligned pair-quantum law in Part II of this document.
'No single workbook readiness formula' vs	Readiness and Forecast are separated from	Do not use readiness language to replace

'Readiness is the primary public-facing individual signal' vs 'user-facing forecast wins'	workbook pair law and versioned as their own contracts. In v1, readiness is self-side/private by default; Forecast is the pair-side guidance product when confidence and sufficiency allow.	compatibility. Do not assume readiness is match-visible.
48,205 vs 48,206 style count conflicts	Counts are normalized by convention: sheet rows include the header row; data rows exclude it.	Never hard-code row counts; validate imported counts at build time and store checksum metadata.
Vague polarity / masculine-feminine shorthand vs polarity v10	Polarity v10 is the only official chemistry subsystem.	Do not score chemistry as masculine minus feminine. Use the ten-axis model only.
Standalone companion modules acting like hidden scoring engines	Companion modules are governed evidence layers with caps.	They may enrich, explain, or suppress. They may not invent or overwrite score law.

4. Locked build principles

Principle	Production meaning
Deterministic first	Every visible score must be reproducible from stored answers, lookup tables, versioned formulas, and auditable packets.
Event-sourced truth	Raw answers and raw events remain authoritative. Inferences, narratives, and forecasts sit on top of them and never erase them.
Backend-only score law	Clients render; they do not own score math, rule penalties, or risk gating.
Consent by scope	Assessment AI, Sanctuary AI, chat analysis, match explanations, couple-shared AI, and training export are separate scopes.
Lane discipline	Fast state, medium pattern, slow identity, and frozen audit/version keys are different classes and may not be collapsed into one blob.
Exposure discipline	Users do not get raw partner risk fields. Matches do not get raw shadow or safety flags. Couple mode does not bypass consent.
No monetized distortion	Coins, badges, vanity codes, premium tiers, and social layers do not affect score, rank, or confidence.
Graceful degradation	If AI, telemetry, or optional runtime layers fail, the deterministic pair engine still returns the official score cleanly.

5. Counting convention, versioning, and import discipline

The uploaded blueprint package mixed sheet dimensions with data-row counts. This revision fixes that by making the counting rule explicit.

- Sheet rows = physical workbook sheet row count, including the header row.
- Data rows = actual importable records, excluding the header row.
- Matrix dimensions are expressed as usable data dimensions, for example 10×10 or 6×6, not as sheet row counts.
- Every import must carry: workbook_version, sheet_name, sheet_rows, data_rows, imported_at, source_checksum, import_job_id, and score_contract_version.
- No service may use a magic number where an import-time validation can provide the truth instead.

Normalized workbook counting convention

Sheet	Sheet rows	Data rows / usable dimensions	Use in code
AnswerBank	311	310 scored answer rows	Import as versioned answer registry; backend-owned.
AnswerPairQuantum	48,206	48,205 pair rows	Import as lookup artifact; never recompute ad hoc in the client.
ArchetypeMatrix	11	10×10 matrix	Load exact fit values.
ShadowMatrix	7	6×6 matrix	Load exact stability/risk values.
Polarity_Axes_v10	11	10 axes	Treat ten axes as official chemistry dimensions.
Polarity_RoleSeeds_v10	6	5 role seeds	Use as scoring/content anchors, not public destiny labels.
Polarity_QuestionMap_v10	33	32 mapped prompts	Attach to polarity question metadata.
Polarity_AnswerBank_v10	161	160 polarity rows	Import like AnswerBank.
Polarity_PairEngine_v10	22	21 data rows / formula registry	Use the exact chemistry formulas and weights.

Why this matters: the merged source file showed both 48,205 and 48,206. Both can be true once the header-row convention is declared. Build code must validate imported counts instead of arguing with a document.

Part II — Canonical scoring engine and workbook import law

6. End-to-end product architecture

Stage	What happens	Primary owner	Output
Onboarding	Create user, RS code, consent profile, demographics, astrology/numerology keys, visibility settings, dating-intent fields.	Auth / profile service	User identity packet
Assessment	Collect 60 answers: 50 core + 10 shadow. Persist answer_id, phase, scoring_version, timestamps.	Assessment service	Versioned answer ledger
Blueprinting	Aggregate answer-row metrics into per-user component profiles and classification outputs.	Scoring service	Self blueprint outputs
Matching	Assemble pair packet, look up pair rows, apply gating, penalties, and overlays, clamp FinalNorm, return 350–850 score and diagnostics.	Matching + scoring services	Compatibility payload
Access	Apply direct-message vs Knock rules according to score/tier and policy.	Matching + messaging	Access decision
Sanctuary	Capture reflections, emotional-state changes, rituals, and governed self-work telemetry.	Sanctuary services	State and reflection artifacts
Relationship mode	Create couple container, shared tools, pulse checks, repair flows, and shared/private rooms.	Relationship service	Relationship container
Calibration	Capture outcomes, drift, repair, messaging cadence, and training labels.	Event ledger + feature store	Calibration datasets

7. Workbook registry and import ownership

Sheet	Role	Primary owner	Implementation note
Schema	Canonical component weights, gate law, penalties, output rules	Backend + QA	Hard-code only through versioned score config generated from the workbook.
AnswerBank	Atomic answer vectors, labels, norms, weights, style metadata	Backend + content ops	Versioned import table; never freehand edit in production.
AnswerPairQuantum	Precomputed pairwise interaction metrics, risk, penalty, pair_norm	Backend + data	Materialize as lookup table or service artifact. Client-side recreation is forbidden.

MatchHarness	Human-readable calculation order and QA fixtures	Backend + QA	Mirror logic in automated tests and frozen fixtures.
ArchetypeMatrix	Archetype pair fit values	Backend	Exact lookup values, not approximate re-creations.
ShadowMatrix	Shadow interaction stability/risk values	Backend	Exact lookup values, hidden from match-facing exposure.
ZodiacLookup / NumerologyLookup	Cosmic overlay norms	Backend + data	Overlays are additive and capped; they never rescue weak core fit.
Polarity package	Chemistry dimensions, role seeds, question map, polarity answer bank, pair formulas	Backend + QA + content	This is the official chemistry subsystem; all older polarity shorthand is retired.
FeedbackData	Outcome telemetry schema for calibration	Data + ML	Implement capture now even if model tuning comes later.

8. Canonical compatibility score law

This is the only official production formula for CompatibilityScore in this blueprint.

Layer	Component	Weight	Workbook field / meaning
Behavioral core	Archetype interaction	16%	archetype_fit
Behavioral core	Communication fit	10%	comm_fit
Behavioral core	Regulation fit	10%	reg_fit
Behavioral core	Love-language fit	6%	love_fit
Behavioral core	Readiness alignment	10%	readiness_fit
Friction / stability	Shadow interaction	9%	shadow_fit
Friction / stability	Integrity burden inverse	7%	integrity_fit = 1 - integrity_burden
Friction / stability	Cluster contradiction inverse	4%	cluster_fit = 1 - cluster_burden
Friction / stability	Pace / attachment mismatch inverse	3%	pace_fit = 1 - pace_mismatch_norm
Polarity / chemistry	Polarity fit	8%	polarity_fit
Polarity / chemistry	Pursuit / receptivity fit	5%	pursuit_fit
Cosmic overlay	Zodiac	7%	zodiac_norm
Cosmic overlay	Numerology	5%	numerology_norm

```
BehavioralComponent = 0.16*archetype_fit + 0.10*comm_fit + 0.10*reg_fit + 0.06*love_fit + 0.10*readiness_fit
StabilityComponent  = 0.09*shadow_fit + 0.07*integrity_fit + 0.04*cluster_fit + 0.03*pace_fit
ChemistryComponent  = 0.08*polarity_fit + 0.05*pursuit_fit
base_0_88           = BehavioralComponent + StabilityComponent + ChemistryComponent
gate                = 1 - (0.55*risk_avg + 0.20*risk_max)
pair_norm_0_1       = max(0, min(1, (base_0_88 * gate) - rule_penalty))
```

```
CoreNorm          = weighted average of pair_norm_0_1 across selected question pairs
CosmicOverlay      = 0.07*ZodiacNorm + 0.05*NumerologyNorm
FinalNorm         = clamp(CoreNorm + CosmicOverlay, 0, 1)
CompatibilityScore  = 350 + 500*FinalNorm
```

Operational meaning: the engine does not simply sum answer scores. Each selected answer is a vector. Each pair of answers maps to a precomputed interaction row. Risk gating and rule penalties keep unstable combinations from scoring too high. Cosmic overlays can refine the result, but cannot dominate it.

Deprecated formula notice: any alternate compatibility formula embedded elsewhere in the merged source package is non-authoritative and must not be implemented in production.

9. MatchHarness runtime order — exact build sequence

Step	Code must do	Do not do this
1. Normalize inputs	Collect selected answer IDs plus gender, sun-sign, and life-path keys for both users.	Do not score from raw answer text or incomplete identity keys.
2. Build pair keys	Sort answer IDs into ans_low|ans_high pair_key for every comparison.	Do not make the join order-sensitive.
3. Lookup pair rows	Fetch component fits, gate, rule_penalty, and pair_norm from AnswerPairQuantum.	Do not recalculate pair metrics ad hoc in the client.
4. Apply question weighting	Multiply by question_weight_v3 from AnswerBank / harness config.	Do not treat all questions equally.
5. Aggregate	Compute CoreNorm, StabilityAvg, ChemistryAvg, and diagnostic drivers as weighted averages.	Do not average without weights.
6. Apply overlays	Add zodiac_norm and numerology_norm using the 7% / 5% weights.	Do not let overlays dominate weak core fit.
7. Clamp and convert	Clamp FinalNorm to [0,1] and convert to the 350–850 public scale.	Do not return values outside the public scale.
8. Tier and explain	Return score, tier, subcomponents, and translated drivers.	Do not expose raw shadow severity or hidden safety/risk fields to matches.

10. AnswerBank, AnswerPairQuantum, matrices, and overlays

AnswerBank is the atomic feature registry. AnswerPairQuantum is the canonical pair lookup artifact. Matrices and overlays are controlled lookup layers. All four are backend-owned data products, not front-end conveniences.

Artifact	What it stores	How developers use it
AnswerBank	answer_id, question metadata, component norms, archetype / shadow tags, pacing, attachment, integrity, forecast, style labels, bands	Render answers in the client; score on the backend; store per-user aggregate rollups.
AnswerPairQuantum	pair_key, answer ids, component fits, forecast_fit, risk_avg, risk_max, gate, rule_penalty, pair_norm	Load during pair computation; preserve exact values for QA and audit.
ArchetypeMatrix / ShadowMatrix	Controlled 10×10 and 6×6 interaction tables	Load exact matrix values into config or lookup tables.

ZodiacLookup / NumerologyLookup	Cosmic overlay norms	Apply only after core aggregation and only through the locked overlay weights.

11. Polarity v10 — official chemistry subsystem

Polarity is not vibe text. It is the official chemistry subsystem inside the 13% chemistry layer, and it is scored as a ten-axis fit model rather than a masculine-minus-feminine shorthand.

Axis	Fit type	Weight inside chemistry	Why it matters
Initiation	Complement	18%	Predicts stall versus motion.
Pursuit preference	Complement	14%	Captures chase rhythm and mutual passivity risk.
Leadership comfort	Complement	12%	Tests who can hold direction without collapse or backlash.
Surrender safety	Similarity	12%	Attraction degrades when closeness feels dangerous.
Calm receptivity	Similarity	12%	Separates grounded chemistry from nervous-system activation.
Tension attraction	Balance @ 0.45	10%	Prevents drama addiction from masquerading as chemistry.
Structure preference	Similarity	8%	Measures desired relational shape.
Energetic flexibility	Similarity	6%	Repair becomes easier when roles can flex.
Validation dependency	Balance @ 0.40	4%	Flags chemistry inflated by approval hunger.
Erotic-safety alignment	Similarity	4%	Tests whether desire and safety reinforce each other.

PolarityDimensionScore_0_100 = 100 × weighted sum of ten axis fit scores

AttractionPatternNorm = 0.28*surrender_safety_fit + 0.22*calm_receptivity_fit + 0.22*erotic_safety_alignment_fit + 0.16*tension_attraction_fit + 0.12*pursuit_fit

ChemistryLayerContribution = (0.08 × PolarityDimensionScore_0_100/100) + (0.05 × AttractionPatternNorm)

Role seeds (MS, Mm, Fa, Fr, Dr) remain internal scoring and content anchors. They are not permanent public identity boxes.

Protection rule: chemistry may elevate or soften a match, but it may never wipe out high risk from integrity burden, contradiction burden, severe pace mismatch, or shadow conflict.

12. Output contract for compatibility

Field	Type	Visibility	Rule
CompatibilityScore	int (350–850)	Match-visible	Primary ranking score.
FinalNorm / CoreNorm	0–1 float	Internal; may be visible to self/admin	Return for diagnostics and QA.

StabilityAvg / ChemistryAvg / overlay norms	0–1 float	Internal; selectively translated	Never expose raw hidden risk fields to matches.
Tier	enum	Match-visible	Thresholds are versioned config, not hard-coded in this document.
Top positive drivers / top friction drivers	translated list	User/match-safe only	Translate; do not dump raw internal safety fields.
Scoring version	string	Internal + admin	Every score must be reproducible.

Part III — Readiness, forecast, clusters, drift, and shadow activation

13. Readiness and Relationship Forecast — corrected doctrine

The workbook defines the pair-quantum compatibility engine. It does not, by itself, define a single public-ready individual readiness formula. The merged source package later introduced separate readiness and forecast contracts. This revision reconciles that conflict instead of pretending it never happened.

Product	What it is	Authoritative source in this package	Public exposure stance in this revision
Compatibility	350–850 pair ranking score	Workbook-aligned pair-quantum law	Primary match-ranking score and access signal.
Readiness	Individual 0–100 relational capacity signal	Separate readiness contract, versioned outside workbook pair law	Self-side/private by default. Never match-visible. May be surfaced later as a self product.
Relationship Forecast	Governed pair-state outlook family	Separate forecast contract	Pair-side guidance product when sufficiency and confidence allow. Does not replace compatibility.

Why this decision is the least fragile: it preserves the workbook-first doctrine, honors the later readiness/forecast contracts as real engineering work, and avoids forcing a non-workbook formula into the core compatibility engine.

14. Readiness contract — separate and versioned

If Readiness is implemented in v1, it must be treated as a separate versioned contract. It is not part of the workbook compatibility formula and must never be described as if it were.

```
readiness_base_01    = (0.34 * AF) + (0.18 * CI) + (0.22 * BR) + (0.16 * CS) + (0.10 * RR)
readiness_penalty_01 = contradiction_penalty + shadow_penalty + drift_penalty + acute_state_penalty
readiness_final_01   = clamp(readiness_base_01 - readiness_penalty_01, 0.00, 1.00)
readiness_score      = round(readiness_final_01 * 100)
```

Component families: AF = assessment foundation, CI = consistency integrity, BR = behavioral regulation, CS = companion stability, RR = reflection and repair capacity.

Exposure rule in this revision: readiness is self-only unless product policy later changes through versioned approval. Never expose another user's readiness to a match.

15. Relationship Forecast contract — pair-state guidance, not prophecy

Forecast is a governed pair output family. It is not fate, it is not a monetized shame score, and it is not a replacement for compatibility.

```
pair_stability_base_01 = (0.40 * compat_norm) + (0.20 * pair_rhythm_norm) + (0.20 * repair_norm) + (0.10 * conflict_norm) + (0.10 * reciprocity_norm)
pair_stability_penalty_01 = (0.30 * shadow_load_norm) + (0.25 * drift_load_norm) + (0.20 * acute_pressure_norm) + (0.25 * contradiction_load_norm)
forecast_pair_stability = round(clamp(pair_stability_base_01 - pair_stability_penalty_01, 0.00, 1.00) * 100)
```

Mandatory forecast outputs should include pair stability outlook, rupture risk, repair capacity, reciprocity health, pacing alignment, shadow pressure index, drift pressure index, and forecast confidence.

Visibility rule: Forecast requires pair sufficiency plus confidence. Low-confidence forecast does not become public-facing certainty.

16. Clusters — governed contradiction and burden engine

Clusters are not personality stickers. They are governed rule objects with explicit inputs, severity, evidence standards, write-back targets, and decay logic.

Severity	Meaning	Default effect	Visibility posture
S1	Informational pattern	Log only or soft reflection	Internal only by default
S2	Mild tension	Light readiness confidence drag or prompt	Softened self language allowed if confidence is medium or higher
S3	Active contradiction / instability	Moderate penalty or pair caution	Self-facing translation only; no raw partner exposure
S4	Repeating destabilizer	Strong readiness drag, forecast caution, AI restraint	Mostly internal; derived self prompts only
S5	Critical pattern	Suppression, gating, safety routing	Internal only except safe intervention copy

Launch-governed clusters: Vulnerability Loop, Control Avoidance, Push-Pull, Repair Gap, Slow Drift, and Readiness Overstatement. Support and watch clusters may feed them but do not bypass the severity ladder or evidence rules.

Core governed cluster family

Cluster	Primary scope	Why it exists	Typical sources
Vulnerability Loop	Individual	User wants closeness but retracts once it becomes emotionally real.	Assessment, messaging, Sanctuary
Control Avoidance	Individual	User preserves control by delaying clarity, staying vague, or resisting structure.	Assessment, conflict events, messaging
Push-Pull	Hybrid	Pursue-withdraw rhythm inflates chemistry and hurts stability.	Assessment, polarity, messaging cadence
Repair Gap	Pair	Measures the distance between caring and actually repairing.	Assessment, conflict/repair telemetry, reflections
Slow Drift	Hybrid	Separates temporary distance from meaningful erosion of reciprocity and upkeep.	Assessment, DriftEligibility, messaging, relationship rituals
Readiness Overstatement	Individual	Protects the self-readiness surface from polished language unsupported by behavior.	Assessment, behavior deltas, forecast write-backs

Support/watch cluster families worth keeping in the rule engine: Reciprocity Imbalance, Intent Mismatch, Pace Mismatch, Integrity Strain, Attachment Spike, Boundary Fog, Surrender Safety Deficit, Structure Collision, Tension Bonding, Validation Dependency, Erotic-Safety Split, Forecast Drag, and the six shadow-activation families.

17. Drift — medium-lane governor, not identity roulette

Drift measures divergence between declared alignment, baseline pattern, and observed relational behavior over time. It is conservative by design.

- Drift may influence readiness or forecast only after DriftEligibility gates pass.
- Slow drift is off by default for archetype changes in v1. Micro-drift may update narrow allowed traits under caps.
- Recent instability should create provisional watch states, not instant identity rewrites.
- A recovering drift state should restore confidence through governed recovery rules, not a one-good-day rebound.

Drift state	Meaning	Effect
none	No meaningful drift evidence yet	No action
provisional_watch	Soft movement pressure exists but gates do not fully pass	Internal watch + prompting only
hold_unstable	System is explicitly preventing slow drift because recent instability is too high	Reduce confidence; do not apply slow trait changes
eligible_micro_drift	Micro-drift may update narrow allowed traits	Capped update path
eligible_slow_drift	All gates pass and confidence is sufficient	Slow drift may update allowed slow traits only through governed job
active_slow_drift / cooldown	An update has been applied and churn is temporarily blocked	Audit log + cooldown

18. Shadow Activation — runtime companion contract

Shadow type from the assessment is not the same as runtime shadow activation. Runtime activation requires governed evidence in a valid time window and never rewrites baseline identity from one volatile sequence.

Shadow activation family	What it tracks	Evidence class	Hard rule
Manipulator	Strategic guilt, leverage, baiting, engineered emotional positioning	Shadow items + messaging + moderation events	AI may read governed outputs; AI may not mint activations on its own.
Stonewaller	Punitive silence, closure refusal, emotional lockout	Shadow items + messaging + conflict events	Pair-facing exposure remains translated and safety-gated.
Love Bomber	Intensity spikes that exceed stability or follow-through	Shadow items + ramp telemetry + drift aftermath	High heat does not rescue low integrity or high risk.
Chameleon	Adaptive self-erasure, mirroring, identity drift for safety/approval	Shadow items + tone shifts + reflections	Requires corroboration, not just one flexible interaction.
Scorekeeper	Ledger-style relating, hidden resentment, conditional giving	Shadow items + reciprocity events + couple conflict logs	Translate softly; keep raw label internal.
Self-Saboteur	Pre-emptive withdrawal, collapse, self-blame before reality is clarified	Shadow items + retreat events + grief/attachment strain	Do not confuse pain spikes with permanent trait truth.

Use runtime shadow activation to influence forecast, interventions, and AI restraint. Do not use it to overwrite workbook answer vectors or baseline shadow identity.

Part IV — Runtime architecture, storage, APIs, and eventing

19. Service ownership

Service / layer	Owns	Reads	Writes
Gateway / API	Auth, validation, throttling, packet routing, policy checks	Consent state, schema registry	Request logs, decision tokens
Scoring service	Workbook law, self aggregates, harness parity	AnswerBank, user answers, imported config	Self blueprint outputs, audit rows
Matching service	Pair packet assembly, compatibility compute, tiering, access decisions	Two self packets, AnswerPairQuantum, matrices, overlays	Pair outputs and diagnostics
Cluster rule engine	Contradiction, burden, severity, write-back policy	Assessment, runtime events, pair outputs	Cluster hits and translated prompts
Drift service	Eligibility gates, drift states, reason codes	Event ledger, rollups, cluster outputs	Drift state objects
Shadow activation service	Runtime activation, decay, downstream packets	Event labels, shadow baseline, conflict windows	Activation artifacts
Sanctuary services	Rituals, reflections, self-state modules	Consent, state, blueprint context	Sanctuary logs, state deltas
Genome extraction / AI layer	Governed feature extraction, narratives, coaching	Allowed packets only	Feature candidates, AI insights, no raw truth overwrite
Event ledger	Append-only event record	All domain events	Immutable event rows
Feature store	Versioned ML features with freshness and provenance	Events, outputs, approved signals	Feature rows and training packets

20. Sync vs async runtime lanes

Lane	Work items	Rule
Synchronous	Assessment submission, deterministic pair scoring, access gating, basic score explanation	Needed for product trust and responsiveness.
Near-real-time	Sanctuary reflections, fast-state mood updates, short-form AI guidance, messaging coaching	Queue-backed; should not block core score return.
Asynchronous	Bulk precompute, drift scans, outcome aggregation, feature refresh, long reports, exports, training jobs	Heavy work belongs in workers, not in the user's tap path.

21. Canonical request flows

Assessment completion:

1. Client submits answers and allowed demographics.
2. Gateway validates schema and consent scopes.

3. Scoring service writes user_answers and computes self aggregates.
4. Blueprint outputs are persisted with scoring_version and source checksum.
5. Optional async jobs enqueue reflections, feature refresh, and report generation.

On-demand match compute:

6. Gateway assembles pair packet from two self packets plus policy mask.
7. Matching service reads AnswerPairQuantum, matrices, overlays, and rule engine context.
8. Service computes CompatibilityScore, subcomponents, tier, and access decision.
9. Report layer filters the payload according to exposure rules before returning it.

Sanctuary / AI reflection:

10. Gateway requests policy decision token first.
11. Allowed context is assembled into a Reflection Packet.
12. Genome extraction and AI layers return self-only narratives or proposals.
13. All saved signals are stored separately from source truth with evidence band and provenance.

22. Must-store objects and schema inventory

Domain	Core tables / objects	Why they exist
Identity & consent	users, consent_state, visibility settings, entitlement records	Identity, privacy, and AI gating.
Assessment & imports	assessment_sessions, questions, answer_bank, answer_pair_quantum, matrices, overlays, polarity tables	Scoring authority and reproducibility.
User outputs	user_answers, user_profile_aggregates, self blueprint outputs, readiness records (if enabled)	Self reports, rescore auditability, feature generation.
Pair outputs	pair_computations, pair_diagnostics, forecast records, access decisions	Match outcomes, explainability, routing.
Clusters & companion runtime	cluster_hits, drift_states, shadow_activation, state packets	Governed runtime interpretation without corrupting core law.
Observational layer	event_ledger, user_state_daily, interaction_reflections, conflict_episodes, repair_events, boundary_events, trigger_events, pair_rhythm_daily	Live behavior, state, and calibration truth.
Sanctuary & AI	sanctuary_logs, genome_features, narrative_memory, ai_insights, packet_outbox, qm_* tables	Governed enrichment, continuity, and ML-ready features.
Relationship mode & social	relationships, couple_rooms, knock records, conversations, messages, economy/badge tables	Product flow and later-stage modules.

23. Packet builder and feature store contract

All non-trivial services should read versioned packets rather than reach directly into arbitrary tables.

Packet	Required contents	Primary consumers
Self Packet	User identifiers, consent scope, demographics, latest assessment outputs, archetype / shadow baselines, polarity	Scoring, reports, AI explanation

	vector, visibility mask	
Pair Packet	Two Self Packets + pair rules + allowed runtime context + policy mask	Matching, forecast, reports
Reflection Packet	Allowed self fields, fast state, selected genome features, explanation boundaries, prohibited fields list	Sanctuary, Quantum Mirror, AI reflection
Training Packet	Feature rows, labels, freshness metadata, confidence weights, provenance ids, exclusion flags	Offline ML and calibration

Every derived feature needs: feature_key, value, freshness window, evidence band, confidence score, consent scope, provenance ids, schema_version, scoring_version, taxonomy_version, and overwrite policy.

24. API surface — minimum build-ready endpoints

Endpoint	Owned by	Returns	Notes
POST /assessment/submit	Gateway → scoring	Self blueprint payload	Primary completion route
GET /profile/score	Gateway	Public-safe self payload	Read-only, cached safely
POST /match/compute	Gateway → matching	Match preview or full report	Checks unlock + consent + exposure policy
POST /sanctuary/reflect	Gateway → Sanctuary / AI	Reflection output	Self-only route
POST /messages/ai-insight	Gateway → AI	Coaching output	Opt-in only; no raw partner exposure
GET /packets/self	Packet builder	Self packet JSON	Internal/admin use
GET /packets/pair	Packet builder	Pair packet JSON	Internal/admin use
POST /training/export	Packet builder + feature store	Dataset manifest	Restricted internal route

25. Failure handling and fallback logic

- Policy denial returns a restricted response with no partial leakage.
- If AI is unavailable, deterministic scores still return; only the optional narrative layer degrades.
- If polarity data is incomplete, compute what is available and mark chemistry completeness honestly.
- If cosmic keys are missing, default the overlay layer safely without blocking core scoring.
- If pair-level dual consent is absent, deterministic match law may still run where policy allows, but pair-level AI insights must be blocked.
- Malformed packets are rejected at the gateway; downstream services are forbidden from guessing missing fields.

Part V — Sanctuary, Quantum Mirror, relationship mode, and governed enrichment

26. Human Emotion Genome and companion-system doctrine

The Human Emotion Genome is the governed feature family that captures tone, phrasing, behavior patterns, narrative themes, trigger classes, boundary signals, regulation markers, and outcome telemetry. It enriches the system; it does not overwrite slow identity recklessly.

Signal family	Typical examples	Lane	Used for
Tone profile	warm open, dry detached, overexplaining, self-blaming	Medium	AI explanation and pattern summaries
Phrasing / stance	joking deflection, qualifier-heavy, mind-reading, growth ownership	Medium	Narrative extraction and coaching
Behavior patterns	pursuer, distancer, appeaser, scorekeeper, repair-oriented	Medium	Forecasting and cluster support
Narrative themes	abandonment fear, overfunctioning, trust fracture, safety craving	Medium	Pattern checks and history entries
Trigger classes	delay, cold tone, mixed signals, pressure, secrecy, ambiguity	Fast / medium	State routing and trigger mapping
Boundary / need signals	space need, transparency need, reassurance preference, pace protection	Medium	Communication guidance and pair context
Regulation markers	shutdown, protest, numbing, self-soothing, restoration ritual	Fast / medium	Companion modules and forecast context
Outcome telemetry	follow-through, repair completion, ghosting, improvement trend	Slow	Calibration and ML labels

Influence cap rule: messaging, Sanctuary, grief, anger, ExsAnonymous, and games may generate governed features and recommendations. They do not get to silently mutate canonical score law or slow identity baselines from one dramatic session.

27. Sanctuary — retained and governed

Module	Job	Write-back class	Key rule
Stillness Mode	Regulation and downshift	Fast-state only	Useful for immediate support; does not rewrite stable identity.
Emotional Mirror	Prompted reflection and self-translation	Fast-state + approved insight	Can save structured signals only within consent and evidence rules.
Whisper Window	Unsent expression / message drafting	Ephemeral by default	Good for relief and clarity; keep partner exposure blocked.

Archive of Self	Longitudinal journal store	Versioned narrative memory	Raw writing and extracted signals must stay separate.
Alignment Compass	State tracking and trend display	Fast-state + rollups	Feeds drift and calibration only through governed paths.
Micro Rituals / Truth Signal	Short interventions	Fast-state	Intervention success can inform later confidence, not pair score directly.
Ghost Mode / Last Light	Recovery, grief, privacy-preserving self-work	Protected state	Do not misread hidden self-work as relational drift.
Polarity Test	Standalone chemistry/axis surface	Profile enrichment	Uses the same polarity v10 package as the core engine.

28. Quantum Mirror — premium AI layer, governed not magical

Quantum Mirror is kept inside the package as the premium private intelligence layer. It reads allowed Genesis OS context, asks better follow-up questions, extracts governed signals from narrative text, and returns evidence-banded guidance. It is not the formula owner.

Quantum Mirror mode	Why user enters	Primary inputs	Output
Score Deep Dive	Explain scores, archetypes, clusters, or match logic	Score outputs, controlled drivers, permitted history	Human-readable explanation
Past Relationship Decode	Capture pattern history and lessons	User narrative, approved memory scope	History entries + themes + reflections
Message Studio	Draft or interpret a message	Current state, match context, preferences	Drafts, tone options, caution notes
After the Date / Conflict	Interpret a recent interaction	Event context, state, perception-reality gap	Grounded reflection + follow-up prompts
Pattern Check	Ask why the same thing keeps happening	Longitudinal trends and validated themes	Pattern summary + interventions
Couple Reflection	Dual-consent shared review	Dual profiles + shared events + shared scope	Joint summary, no unilateral leakage

- AI may explain, propose, moderate, and draft. AI may not directly overwrite archetype, baseline shadow, baseline polarity, or CompatibilityScore.
- Every saved signal requires provenance, evidence band, time scope, extractor version, and consent scope.
- Conversation memory, structured profile signals, and moderation evidence are different stores with different retention rules.

29. Messaging, Knock, relationship mode, and product modules

Module	Keep?	Use in this blueprint	Guardrail
Knock system	Yes	Intentional request flow when direct messaging is not open	Thresholds are config-managed; premium does not buy rank or bypass.
Messaging telemetry	Yes	Captures initiation balance, response time, repair attempts, silence windows, and	Requires consent for content-level AI analysis; event metrics may still be logged where

		reciprocity	policy allows.
Relationship mode	Yes	Creates shared container, couple handle, pulse checks, repair modules, and shared/private rooms	Dual consent required for shared AI and shared write-backs.
Symphonic Match	Yes, later phase	Downstream pair artifact driven by real pair data	Never allowed to replace compatibility or forecast.
Coins / badges / social identity	Yes, as engagement layer	Cosmetics, progression, and community signals	Cannot change score, rank, confidence, or match eligibility.
Relationship Coach / later optional tools	Later phase	Structured analysis and intervention module	Must obey the same packet, consent, and audit rules as other AI surfaces.

30. Relationship mode — shared/private separation law

- Relationship confirmation creates a shared relationship container, not a raw data merge.
- Every shared artifact must declare whether it is shared, partner-private, or system-internal.
- Pulse checks, repair modules, shared Sanctuary rooms, and milestone reflections may write to the relationship container; unilateral private reflections do not become partner-visible by default.
- Breakup / post-link mode must preserve private history, close shared access safely, and route grief/recovery systems without corrupting historical score records.

Part VI — QA, rollout, schema inventory, and implementation warnings

31. Build order and rollout sequence

Phase	Build now	Primary output	Do not wait for
Phase 1 — law stack and imports	Import workbook config tables; create AnswerBank and AnswerPairQuantum stores; implement scoring and matching services; create MatchHarness QA cases; add event ledger and consent model	Deterministic compatibility score, diagnostics, and auditability	AI polish or social extras
Phase 2 — product MVP	Assessment flow, self blueprint, candidate generation, pair compute endpoint, Knock / messaging rules, basic self report, basic pair report	Users can complete assessment and receive scored matches	Full emotional genome depth
Phase 3 — chemistry and explanation	Deploy polarity v10 outputs, chemistry card, translated explanations, top-driver reporting, starter cluster handling	Reports feel intelligent instead of mystical	Advanced model tuning
Phase 4 — Sanctuary and governed AI	Reflections, Whisper Window, state packets, genome extraction, policy-gated AI narratives	Helpful self tools without score corruption	Couple-shared AI depth
Phase 5 — outcome learning	FeedbackData capture, calibration dashboards, feature store, cohort evaluation, score drift monitoring	System starts learning from real outcomes	Autonomous reweighting
Phase 6 — relationship mode and advanced systems	Couple tools, richer drift logic, Symphonic Match, social/economy layers, optional coach modules	Post-match retention and forecasting depth	Anything that compromises core law

32. Deterministic QA and acceptance criteria

- Write at least five frozen MatchHarness scenarios that must exactly match the workbook outputs for score, FinalNorm, CoreNorm, StabilityAvg, ChemistryAvg, ZodiacNorm, and NumerologyNorm.
- Verify pair_key sorting is order-insensitive for every pair row lookup.
- Verify score outputs are identical across environments when imports, versions, and inputs match.
- Verify all public-safe surfaces hide raw shadow severity, hidden safety flags, raw cluster ids, and moderation artifacts.
- Verify degraded mode: missing AI still returns deterministic scores cleanly; missing optional telemetry lowers confidence but does not corrupt score law.
- Verify event-ledger replay can reconstruct derived outputs and packets.
- Verify every displayed score is traceable to score_version, policy version, input window, and source checksum.
- Verify no premium, coins, badges, or vanity identity can affect score, rank, access thresholds, or confidence.
- Verify retention and deletion rules keep raw source truth, extracted signals, and public narratives separate.

33. Non-negotiable implementation warnings

Do not	Why
Do not replace AnswerPairQuantum lookup with a loose front-end formula.	You will lose auditability and create silent score drift.
Do not expose raw shadow, moderation, or safety fields to matches.	That is governance failure dressed as transparency.
Do not let premium or social features affect score, rank, or confidence.	The product becomes a casino and trust dies.
Do not treat role seeds or clusters as public destiny labels.	They are internal scoring and explanation controls, not permanent tattoos.
Do not write AI narrative outputs back into baseline truth without evidence and consent.	One dramatic session should not rewrite a person's file forever.
Do not let one fight, one grief week, or one silent window permanently rewrite archetype, baseline shadow, or polarity profile.	That collapses fast state into slow identity and breaks the whole system.
Do not implement direct client-side score calculations for convenience.	Different clients will drift and your tests will stop meaning anything.

34. Final developer handoff statement

Give developers this document together with the authoritative workbook import package, the import scripts, the MatchHarness fixtures, and the schema registry. Build the pair engine from the workbook-aligned law here. Build the runtime layers from the governed sections here. Do not cherry-pick older formulas because they sound cleaner in a meeting.

If an older Word or PDF fragment disagrees with this document, this document wins. If this document ever disagrees with the imported workbook score law, the workbook wins and this document must be revised.

Appendix A — Controlled labels and canonical enums

Label family	Canonical values kept in this blueprint
Archetypes	Analyzer, Fixer, Icebox, Performer, Phantom Seeker, Quiet Exit, Regulated Grown-Up, Romantic Idealist, Survivor, Translator
Shadow types	Chameleon, Love Bomber, Manipulator, Scorekeeper, Self-Saboteur, Stonewaller
Risk / positive / chemistry bands	Low, Medium, High, Very Low where applicable; keep versioned workbook values
Access style	Guarded, Measured, Open, Selective
Conflict style	Avoidant, Direct, Emotional, Indirect, Logical, Mixed
Repair style	Co-repair, Delay, Re-engage, Suppress
Attachment style	Mixed, Rush-risk, Slow-build
Nervous-system bias	Calm-seeking, Mixed
Update lanes	fast, medium, slow, frozen
Confidence / evidence bands	low, guarded, medium, high, insufficient / A-B-C-D where contract-specific

Appendix B — Cluster catalog summary

Cluster	Family	Primary sources	Why it stays in the package
Vulnerability Loop	Governed contradiction	Assessment + messaging + Sanctuary	Closeness desired, retreat after closeness becomes real.
Control Avoidance	Governed contradiction	Assessment + conflict + messaging	Stability desired, clarity / structure resisted.
Push-Pull	Governed behavioral	Assessment + polarity + cadence	Pursue-withdraw rhythm inflates chemistry and reduces safety.
Repair Gap	Governed conflict	Assessment + repair telemetry + reflections	Caring and actual repair do not line up.
Slow Drift	Governed behavioral	Assessment + DriftEligibility + cadence	Longitudinal erosion of reciprocity and upkeep.
Readiness Overstatement	Governed readiness	Assessment + behavior deltas	Self-readiness surface is more optimistic than behavior supports.
Reciprocity Imbalance	Support / watch	Assessment + messaging + couple mode	One-sided effort or chronic emotional labor asymmetry.
Intent Mismatch	Support / watch	Onboarding + assessment + pair packet	Chemistry exists but direction is structurally mismatched.
Pace Mismatch	Support / watch	Assessment + matching + drift	Misaligned speed for intimacy or commitment.
Integrity Strain	Support / watch	Assessment + shadow items + events	Values, honesty, and observed behavior drift apart.

Attachment Spike	Support / watch	Assessment + messaging + distress windows	Sudden activation under silence or uncertainty.
Boundary Fog	Support / watch	Assessment + Sanctuary + ExsAnonymous	Hints, over-accommodation, or self-erasure replace direct asks.
Surrender Safety Deficit	Chemistry-risk	Assessment + polarity + Sanctuary	Attraction present, but closeness does not feel safe enough to relax into.
Structure Collision	Chemistry-risk	Polarity + intent + couple mode	Pair cannot agree who carries direction or relational shape.
Tension Bonding	Chemistry-risk	Polarity + conflict cadence	Chemistry is being powered by friction, uncertainty, or chaos.
Validation Dependency	Chemistry-risk	Assessment + polarity + messaging	Chemistry is inflated by feeling chosen or mirrored.
Erotic-Safety Split	Chemistry-risk	Polarity + reflections + relationship mode	Desire and safety run on separate rails.
Forecast Drag	Forecast meta-cluster	Pair engine + drift + repair + outcomes	Multiple lighter burdens compound into weaker pair stability.

Appendix C — Sanctuary, Quantum Mirror, and relationship-mode module registry

Module family	Modules kept	Why they matter	Core guardrail
Sanctuary	Stillness, Emotional Mirror, Whisper Window, Archive of Self, Alignment Compass, Micro Rituals, Truth Signal, Ghost Mode, Last Light, Polarity Test	Private self-work, state regulation, reflection, and governed signal capture	No private reflection or grief marker becomes match-visible by default.
Quantum Mirror	Score Deep Dive, Past Relationship Decode, Message Studio, After the Date, After the Conflict, Pattern Check, Couple Reflection	Premium explanation, guidance, structured extraction, continuity	AI reads allowed packets and writes proposals or memories only within consent and evidence rules.
Relationship mode	Couple handle, shared/private rooms, pulse checks, repair modules, shared Sanctuary, breakup / post-link mode	Post-match retention and pair maintenance	Dual consent is mandatory for shared write-backs and shared AI.

Appendix D — Explicitly deprecated formulas, shortcuts, and bad ideas

Deprecated item	Why it is out
Any legacy compatibility formula that is not the workbook-aligned pair law	Multiple generations of the merged source file carried older formulas. Production gets one formula only.
The alternate '0.30*B + 0.22*A + 0.18*(1 - S_risk) + 0.12*P + 0.08*Astro + 0.05*Num' compatibility formula	It conflicts with the workbook-aligned pair-quantum law and was one of the main sources of drift risk.

Client-side pair-score recreation	It destroys parity, auditability, and deterministic QA.
Masculine-minus-feminine chemistry shorthand	Polarity v10 replaced it with the ten-axis chemistry model.
Public match exposure of raw shadow, safety, moderation, or cluster ids	Unsafe, easy to misuse, and contrary to the governance model.
Any feature that lets money alter score, rank, eligibility, or confidence	Hard forbidden by this blueprint.
Using one dramatic chat or reflection to rewrite slow identity	Fast state is not slow identity.

Appendix E — Release checklist

- Workbook imports validate and store counts, versions, and checksums.
- AnswerBank, AnswerPairQuantum, polarity tables, and matrices are all backend-accessible and versioned.
- Five or more frozen MatchHarness scenarios pass with exact parity.
- CompatibilityScore, FinalNorm, CoreNorm, StabilityAvg, ChemistryAvg, ZodiacNorm, NumerologyNorm, tier, and translated driver payloads return from the official service.
- Readiness and Forecast, if enabled, run from separate versioned contracts and do not leak into compatibility law.
- Cluster engine, drift, and shadow activation write back through governed ledgers only.
- Sanctuary and Quantum Mirror are policy-gated and keep source truth, extracted signals, and public narratives separate.
- Relationship mode preserves shared/private separation and dual-consent boundaries.
- No premium, social, badge, coin, or vanity layer can affect ranking, score, confidence, or access thresholds.
- All public-facing explanation copy is safe, non-diagnostic, and versioned.